

EduGenie Scalability & Future Plan

Date	14 July 2026
Team ID	XXXXXX
Project Name	EduGenie – AI-Powered Educational Assistant
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how EduGenie can be scaled to handle larger numbers of learners, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the platform beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address
1	Limited support for concurrent users	Response times may degrade when many learners access the assistant simultaneously	High
2	Limited integration with external learning platforms (LMS, course portals)	Restricts content import and broader learning ecosystem connectivity	High
3	No long-term storage of quiz history or learning progress	Learners cannot track past performance or revisit previous learning paths	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports a limited number of learners on a single server	Deploy on cloud infrastructure (AWS, Google Cloud, Azure) with load balancing and auto-scaling
2	Data Storage	Basic storage for learner queries and generated content	Implement scalable databases such as MongoDB or PostgreSQL with cloud storage support
3	Performance	AI responses (answers, quizzes, summaries) are generated in real-time for each request	Use caching mechanisms, asynchronous processing, and optimized Gemini AI inference pipelines
4	Security	Basic authentication and API security	Implement OAuth, encryption, role-based access control, and secure API gateways

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	Integration with LMS platforms, Google Classroom, and course content repositories	2 Months	Enables richer learning material import and broader subject coverage
Phase 3	AI-powered adaptive quizzes and personalized study reminders	4 Months	Improves self-assessment accuracy and learner engagement
Phase 4	Advanced analytics dashboard, multilingual support, and mobile application	6 Months	Expands accessibility, enhances user experience, and supports global learners